

COMPUTER SCIENCES, A.Y.2024/2025

Laboratory 1

Objectives of the laboratory

- Familiarize yourself with the development and debug environment (Visual Studio Code)
- Run programs already written in Python to become familiar with the mechanism of data acquisition, visualization and execution

If possible, use your own laptop.

PRELIMINARY STEP:

- **If you are using your own laptop**, follow the instructions in the document uploaded to the portal for installing **Visual Studio Code** on your device.
 - **If not**, open the Visual Studio Code application on the laboratory PC;
Do this test even if you have your own laptop; it will be useful for subsequent sessions, if you will use the lab PCs.
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Exercise 1. Using Visual Studio Code, write and run the following program making sure that there are no *errors* or *warnings* during build.

```
x = input("Enter an integer: ");
x = int(x)
y = 3;
z = x/y;
print("x= ",x,"y= ",y,"z= ",z)
```

After you run it, **use the debugger's step-by-step execution functionality, by looking at the value of the x, y, and z variables.**

Exercise 2. Using Visual Studio Code, write and run the following program making sure that there are no *errors* or *warnings* during build.

```
a = input("Enter the value A: ")
b = input("Enter the value B: ")
c = input("Enter the value C: ")
print("A =",a,"B =",b,"C =",c)
```

```

if a>=b:
    if a>=c:
        print("B is the maximum")
    else:
        print("C is the maximum")
else:
    if b>=c:
        print("A is the maximum")
    else:
        print("C is the maximum")

```

After running it, **use the debugger's step-by-step execution functionality, trying different combinations of A, B, C values.** Then, draw a flowchart that solves the same problem (which is: *"find the maximum between A, B and C"*). Run the program above step-by-step, and verify that the flow of execution of the program and of the flowchart are the same, with different input combinations.

Exercise 3. Using Visual Studio Code, write and run the following program making sure that there are no *errors* or *warnings* during build.

```

sum = float(input("enter initial sum: "))
interest = float(input("enter rate: "))
Year = 0
balance = sum
while balance<2*sum:
    year = year + 1
    balance = balance * (1+interest)
print("Your capital will double in", year, 'years, and will be
equal to', balance, 'euro')

```

After running it, **use the debugger's step-by-step execution functionality, trying different combinations of initial sum and interest rate.** Be especially careful about the execution sequence during the iteration. Finally, build a flowchart of the solution from the code (we have seen a very similar example in class).